Worshop Week 2: Answers to questions missed in the solution

Q1. C

110011 = 1x2^5+1x2^4+0x2^3+0x2^2+1x2^1+ 1x2^0=51

Q2. C 35 = 100011

35 / 2 =17 R 1

17 / 2 =8 R 1

8 / 2 =4 R 0

4 / 2 =2 R 0

2 / 2 =1 R 0

1 / 2 =0 R 1

Q3. C

10011101

= 1001 1101

= 9 13

= 9D

Q4. C

1C

= 0001 1100

=00011100 =11100

Q5. B

15 – 17

15 = 0000 1111

17 = 0001 0001

One’s complement of 17 = 1110 1110

-17: complement + 1 = 1110 1110 + 1 = 1110 1111

15 = 0000 1111

-17= 1110 1111

1111 1110

One’s complement of 1111 1110 = 0000 0001

Two’s complement: 0000 0001 + 1 = 0000 0010 = 2

Result is -2.

15-17 = -2

Q5. C

28 – 20

28 = 0001 1100

20 = 0001 0100

Complement of 20 = 1110 1011

-20 = 1110 1011 + 1 = 1110 1100

28 = 0001 1100

-20= 1110 1100

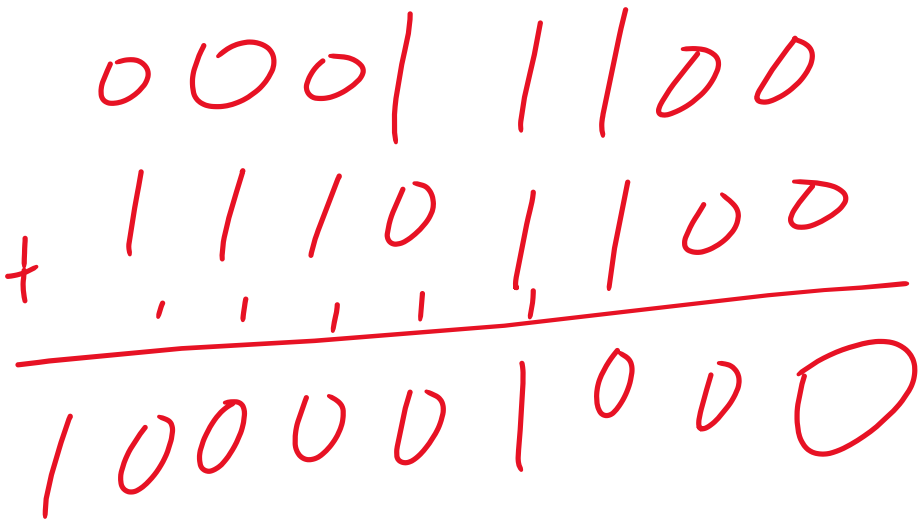
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1 0000 1000 (discard the left most 1, we only have 8 bits)

= 0000 1000

= 8

28+(-20)=8



Q6. C

0111 1111 = 127

Q6. D

1111 0101

One’s complement = 0000 1010

Complement + 1 = 0000 1010 + 1 = 0000 1011 = 11

1111 0101 = -11

Q7

68 65 78 61 64 65 63 69 6D 61 6C = h e x a d e c i m a l

Q8. B

6 bits

Unsigned range: 00 0000 -> 11 1111

Signed range: 10 0000 -> 01 1111 (-32 -> 31)

Q8.C

8 bits

Unsigned range: 0000 0000 -> 1111 1111

Signed range: 1000 0000 -> 0111 1111 (-128 -> 127)

Q9

Please see the source code: MaxIntExample.cpp